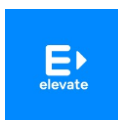


E-commerce Sales Data

Excel Module End Assignment
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Student at Entru Elevate

1. Project Overview

The objective of this project is to analyse Ecommerce sales data using Microsoft Excel and generate meaningful business insights through data cleaning, transformation, analysis, and visualization.

This project demonstrates key Excel skills required for a Data Analyst role:

- Data Cleaning & Validation
- Formulas and Functions
- Pivot Tables & Pivot Charts
- Dashboard Creation
- Business Insights Generation

2. Business Problem

Management wants to understand the following business questions to drive data-driven decisions:

- Which products generate the highest revenue?
- Which regions perform best?
- Yearly sales trends
- Product category performance

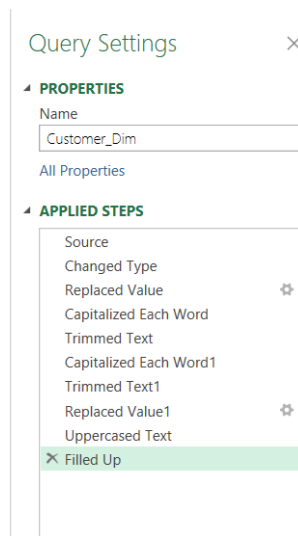
3. Dataset Information

Dataset Name: Ecommerce Sales Dataset | Records: 2000 Transactions

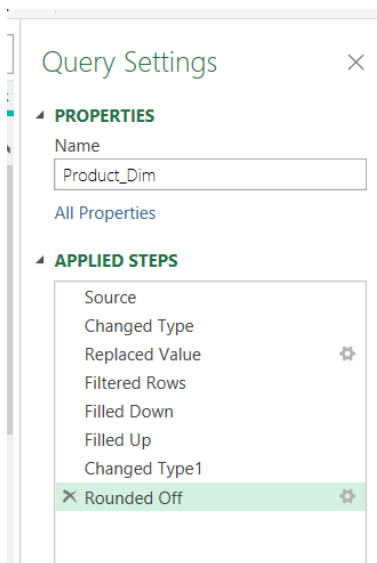
Column Name	Description
Sales ID	Unique Number against each sale
Order Date	Date of Purchase
Customer ID, Name, Age, Gender, City, State, Country	Customer Details
Customer Loyalty level	Denotes Customer Membership
Product ID, Name, Category, Sub-category, Brand	Product Details
Product Cost	Actual cost of the product
Product Stock	Units available
Store ID, Name, Region, City, Type	Store Details
Payment Type	Mode of Payment
Quantity	Units Sold
Unit Price	Price per Unit
Revenue	Quantity * Unit Price
Discount and Discount %	Discount offered

4. Data Cleaning Process

Customer Data- Data transformation in Power Query



Product Data- Data transformation in Power Query



Sales Fact Sheet

Data Imputation for Quantity using If Function:

`=IF(ISBLANK([@Quantity]),AVERAGEIF(E:E,[@[Store_ID]],F:F),[@Quantity])`

=IF(ISBLANK([@Quantity]),AVERAGEIF(E:E,[@Store_ID],F:F),[@Quantity])											
	A	B	C	D	E	F	G	H	I	J	K
	Sales_ID	Order_Date	Customer_ID	Product_ID	Store_ID	Quantity	Quantity New	Unit_Pric	Discoun	Payment_Typ	Total_Amount
1	SALE0	2024-07-07	CUST1409	PROD97	STORE3	1	1	218.04	0.09	Credit Card	198.4164
2	SALE1	2023-12-25	CUST1467	PROD74	STORE17	3	3	104.47	0.27	PayPal	228.7893
3	SALE2	2024-06-03	CUST1056	PROD49	STORE8	4	4	914.05	0.25	Credit Card	2742.15
4	SALE3	2023-11-29	CUST1346	PROD76	STORE8	1	1	921.51	0.16	COD	774.0684
5	SALE4	2024-01-06	CUST1493	PROD59	STORE5	3	3	745.3	0.18	Credit Card	1833.438
6	SALE5	2024-07-26	CUST1116	PROD15	STORE7	3	3	147.01	0.19	COD	357.2343
7	SALE6	2025-07-12	CUST1016	PROD56	STORE4	2	2	785.31	0.15	PayPal	1335.027
8	SALE7	2023-11-18	CUST1152	PROD70	STORE8	4	4	342.39	0.23	PayPal	1054.5612
9	SALE8	2025-05-04	CUST1187	PROD99	STORE5		=IF(ISBLANK([@Quantity]),AVERAGEIF(E:E,[@Store_ID],F:F),[@Quantity])				
10	SALE9	2025-07-31	CUST1464	PROD33	STORE0	2	IF(logical_test, [value_if_true], [value_if_false]) 3 PayPal				
11	SALE10	2024-02-13	CUST1065	PROD24	STORE14	1	1	278.63	0.07	Credit Card	259.1259
12	SALE11	2025-05-24	CUST1320	PROD30	STORE1	4	4	286.03	0.17	Credit Card	863.5344

Data Imputation for Unit Price using Vlookup Function:

=VLOOKUP([@[Product_ID]],Product_Dim_1[[Product_ID]:[Cost]],6,0)

Clipboard		Font		Alignment		Number		Styles		Ce	
=VLOOKUP([@[Product_ID]],Product_Dim_1[[Product_ID]:[Cost]],6,0)											
A	B	C	D	E	F	G	H	I	J	K	L
Sales_ID	Order_Date	Customer_ID	Product_ID	Store_ID	Quantit	Quantity New	Unit_Pric	Discour	Payment_Type	Total_Amount	
SALE19	2024-03-18	CUST1239	PROD60	STORE13	4	4	=VLOOKUP([@[Product_ID]],Product_Dim_1[[Product_ID]:[Cost]],6,0)				
SALE52	2025-08-25	CUST1148	PROD89	STORE6	3	3	VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])				
SALE268	2024-09-20	CUST1406	PROD90	STORE18	3	3	340.14	0.06	Credit Card	315.2478	

Query Settings

NAME: Sales_Fact

ALL PROPERTIES

APPLIED STEPS

- Source
- Changed Type
- Rounded Off
- Rounded Off1
- Rounded Off2
- Removed Columns
- Rounded Off3
- Removed Columns1
- Changed Type1
- Removed Columns2

Combine data using Merge Queries:

Sales_Fact (2) - Power Query Editor

Table: RenameColumns(*Reordered Columns2*,{"Quantity New","Unit Price", "Total Amount"})

Sales_ID	Order_Date	Customer_ID	Customer_Dim.Name	Customer_Dim.Age	Customer_Dim.Gender	Customer
SALE0	07-07-2024	CUST1409	Sarah Smith	55	Male	Kathleenpo
SALE25	17-09-2024	CUST1381	Erika Padilla	50	Male	Andrewslan
SALE72	29-08-2024	CUST1409	Sarah Smith	55	Male	Kathleenpo
SALE434	06-12-2023	CUST1148	Theresa Campbell	42	Female	New Brian
SALE233	11-08-2025	CUST1010	Dominic Page	53	Male	Port Michael
SALE283	13-12-2024	CUST1022	Paul Fox	33	Female	Michaelton
SALE388	03-05-2025	CUST1000	Sheena Bonilla	55	Male	Donaldton
SALE1	25-12-2023	CUST1467	Michael Foley	63	Female	Oneilfurt
SALE459	24-01-2024	CUST1467	Michael Foley	63	Female	Oneilfurt
SALE20	05-07-2025	CUST1144	Justin Cross	62	Female	North Anth
SALE105	15-10-2023	CUST1002	Dr. Benjamin Price	29	Female	Kylebury
SALE301	31-10-2024	CUST1028	Joshua Ryan	27	Female	New David
SALE17	11-12-2024	CUST1079	Molly Gardner	22	Female	Nicoleboros
SALE57	19-06-2024	CUST1467	Michael Foley	63	Female	Oneilfurt
SALE430	02-06-2024	CUST1001	Eugene Nelson	26	Female	Phillipsmou
SALE113	01-05-2024	CUST1467	Michael Foley	63	Female	Oneilfurt
SALE76	25-12-2023	CUST1019	Michael Torres	46	Female	Gonzalezfor
SALE211	30-05-2025	CUST1012	Kenneth Lambert	55	Female	North Kathr
SALE2	03-06-2024	CUST1056	Christopher Sutton	49	Female	Deborahber
SALE3	29-11-2023	CUST1346	John Jordan	58	Female	Port Crystal
SALE30	18-10-2023	CUST1028	Joshua Ryan	27	Female	New David
SALE317	15-06-2025	CUST1346	John Jordan	58	Female	Port Crystal
SALE4	06-01-2024	CUST1493	Amanda Powell	28	Male	South Willi
SALE23	23-06-2024	CUST1368	Miranda King	34	Female	West Math
SALE148	09-08-2025	CUST1493	Amanda Powell	28	Male	South Willi
SALE326	24-09-2025	CUST1187	Dr. Suzanne Bullock	60	Female	Lake Blaket
SALE151	29-10-2024	CUST1022	Paul Fox	33	Female	Michaelton
SALE28	23-09-2025	CUST1148	Theresa Campbell	42	Female	New Brian
SALE5	26-07-2024	CUST1116	Anthony Phillips	18	Female	North Davic
SALE215	05-03-2024	CUST1443	Adam Castro	51	Male	New Donna
SALE244	03-05-2025	CUST1200	Anna Perez	30	Male	South Jerry
SALE315	21-05-2024	CUST1116	Anthony Phillips	18	Female	North Davic

Query Settings

NAME: Sales_Fact (2)

ALL PROPERTIES

APPLIED STEPS

- Source
- Changed Type
- Rounded Off
- Rounded Off1
- Rounded Off2
- Removed Columns
- Rounded Off3
- Removed Columns1
- Changed Type1
- Removed Columns2
- Merged Queries
- Expanded Customer_Dim
- Expanded Product_Dim
- Reordered Columns1
- Merged Queries2
- Expanded Store_Dim
- Reordered Columns2
- Renamed Columns

5. Descriptive Statistics for Unit Price, Total Amount, Actual Discount and Discount %

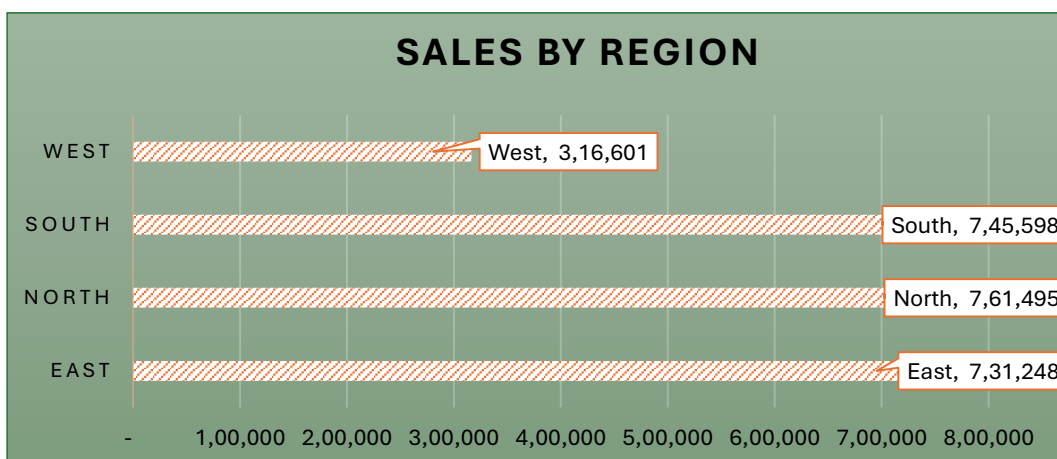
Unit_Price		Total Amount		Actual Discount		Discount %	
Mean	513.72	Mean	1277.47	Mean	192.02	Mean	0.15
Standard Error	6.31	Standard Error	21.25	Standard Error	4.54	Standard Error	0.00
Median	522.43	Median	1009.24	Median	119.55	Median	0.14
Mode	350.12	Mode	700.24	Mode	0.00	Mode	0.10
Standard Deviation	282.22	Standard Deviation	950.30	Standard Deviation	203.15	Standard Deviation	0.10
Sample Variance	79646.89	Sample Variance	903068.31	Sample Variance	41271.75	Sample Variance	0.01
Kurtosis	-1.19	Kurtosis	-0.14	Kurtosis	3.03	Kurtosis	10.85
Skewness	-0.03	Skewness	0.82	Skewness	1.64	Skewness	1.35
Range	979.80	Range	3971.66	Range	1812.71	Range	1.32
Minimum	20.09	Minimum	21.14	Minimum	-372.53	Minimum	-0.32
Maximum	999.89	Maximum	3992.80	Maximum	1440.18	Maximum	1.00
Sum	1027433.52	Sum	2554941.90	Sum	384034.66	Sum	299.14
Count	2000	Count	2000	Count	2000	Count	2000

6. Pivot Table Analysis

Sales by Region:

Region	Revenue
East	₹ 7,31,248
North	₹ 7,61,495
South	₹ 7,45,598
West	₹ 3,16,601

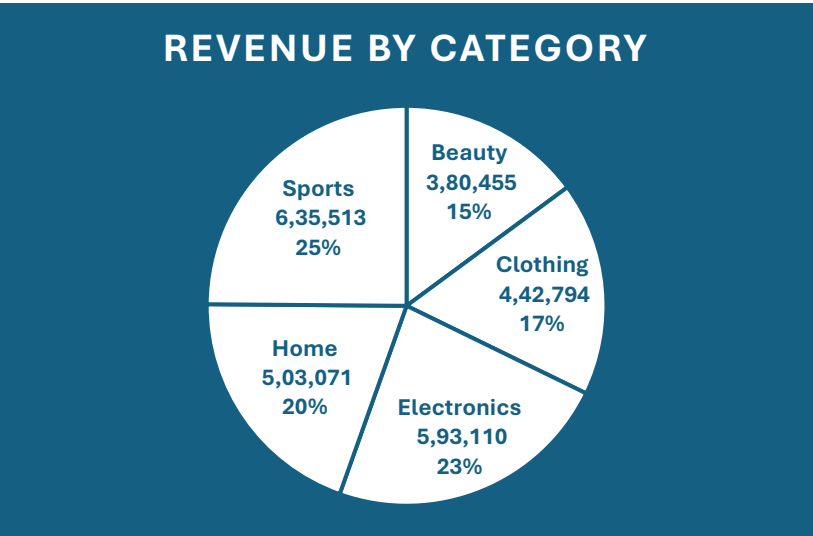
Insight: North Region generated the highest sales revenue among all four regions.



Revenue by Category:

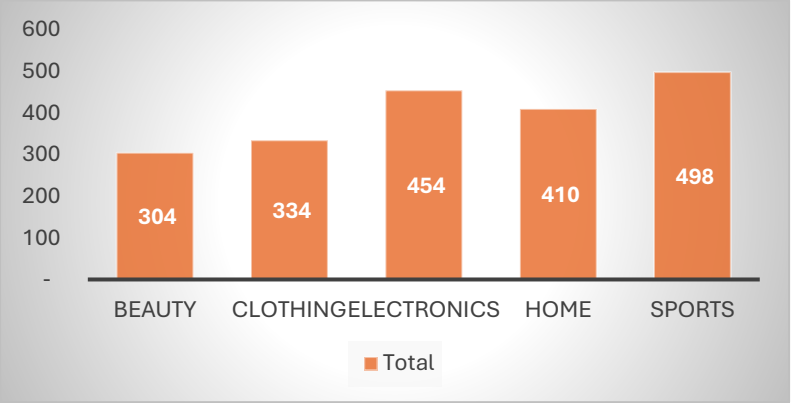
Category	Revenue
Beauty	₹ 3,80,455
Clothing	₹ 4,42,794
Electronics	₹ 5,93,110
Home	₹ 5,03,071
Sports	₹ 6,35,513

Insight: Sports is the top-performing category, contributing nearly 25% of total revenue.

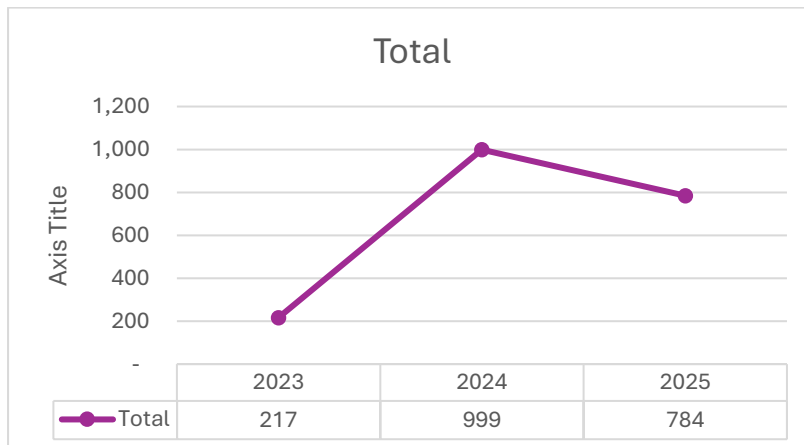


Sales by Category:

Category	Number of Sales
Beauty	304
Clothing	334
Electronics	454
Home	410
Sports	498



Yearly Sales Trend:



7. Visualizations Created

Chart Type	Purpose	Key Insight
Bar Chart	Compare sales across regions	North = Highest sales revenue
Column Chart	Compare category-wise sales	Sports = 24.9 %
Pie Chart	Show category contribution %	Sports ≈ 25% of total revenue
Line Chart	Yearly sales trend	49.9% of Sales in year 2024

8. Key Findings

- ❖ **Massive Growth in 2024:** Sales jumped from 217 in 2023 to 999 in 2024 — a **360% increase**.
- ❖ **Slight Decline in 2025:** Sales dropped to 784, a **21.5% decrease** compared to 2024, though still far above 2023 levels.
- ❖ **Peak Year:** 2024 was the strongest year, contributing half of all sales.
- ❖ **Sustained Growth vs. Volatility:** While 2025 dipped, the overall trend from 2023 → 2025 is still strongly upward.

9. Key Business Insights

- ❖ **Growth Opportunity:** Beauty is underperforming compared to other categories. Targeted promotions or product expansion could lift this segment.
- ❖ **Revenue Concentration:** Sports + Electronics together account for nearly **45% of total revenue**, showing strong dependence on these two categories.
- ❖ **Diversification Need:** Heavy reliance on Sports/Electronics could be risky; boosting Beauty and Clothing would balance the portfolio.
- ❖ **Strategic Focus:** Sports is already strong—investing in sustaining demand here while innovating in Beauty could maximize overall growth.

10. Strategic Takeaways

- **Sports & Electronics:** Protect and sustain these categories—they are your core drivers.
- **Beauty & Clothing:** Need targeted growth strategies (bundles, promotions, premium positioning).
- **Home:** Reliable contributor; could be leveraged as a steady growth category.

11. Skills Demonstrated

✓ Data Cleaning ✓ Data Validation ✓ Conditional Formatting ✓ Excel Functions & Formulas	✓ Dashboard Design ✓ Pivot Tables & Pivot Charts ✓ Data Visualization ✓ Business Insights Generation
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12. Conclusion

This Excel Data Analyst project successfully transformed raw e-commerce sales data into actionable business insights. Using Excel's analytical capabilities — including pivot tables, formulas, dashboards, and visualizations — management can now make informed decisions to improve revenue, profitability, and operational efficiency.